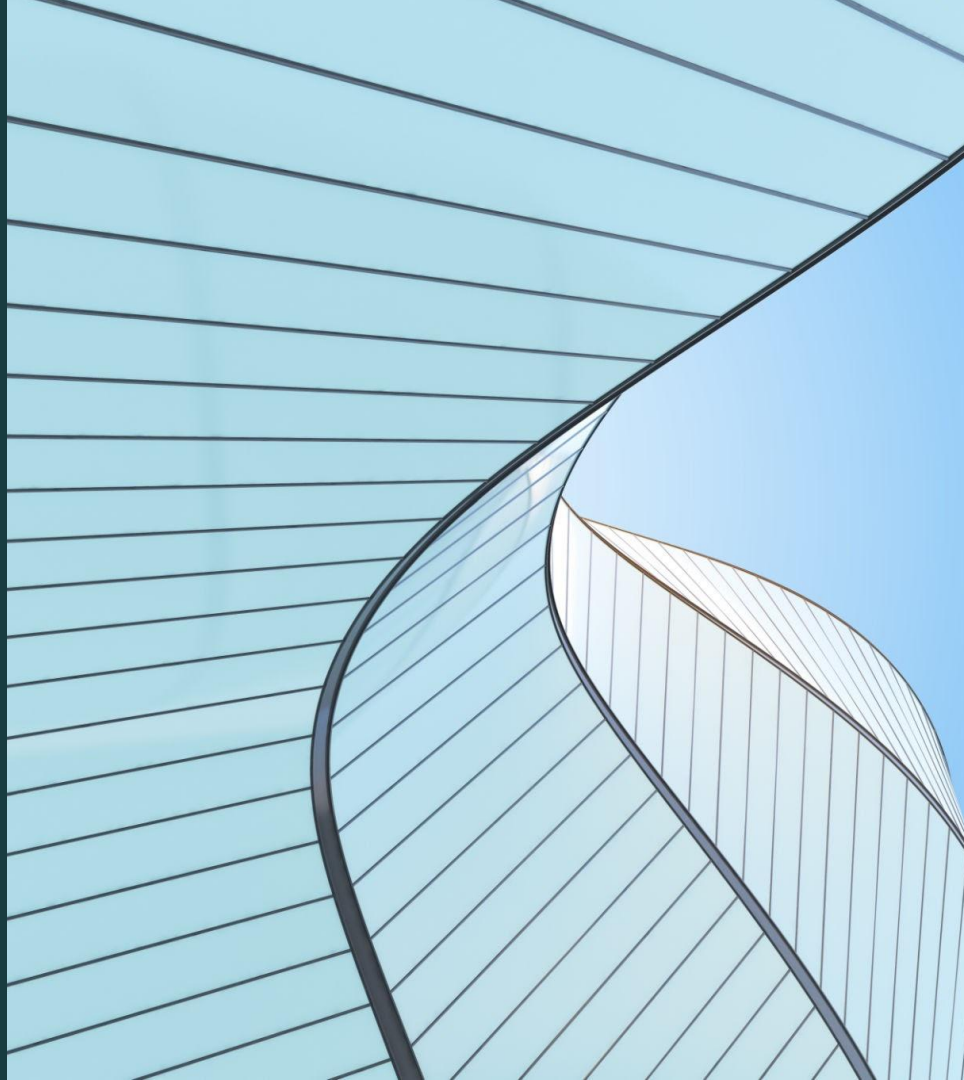
Karl Muller
October 25th, 2026

RL for NYC's Special Natural Area Districts

Special Natural Area Districts (SNADs) are a pressing complaint issue for NYC's DOB and Reinforcement Learning can be used to more efficiently triage & target these issues.



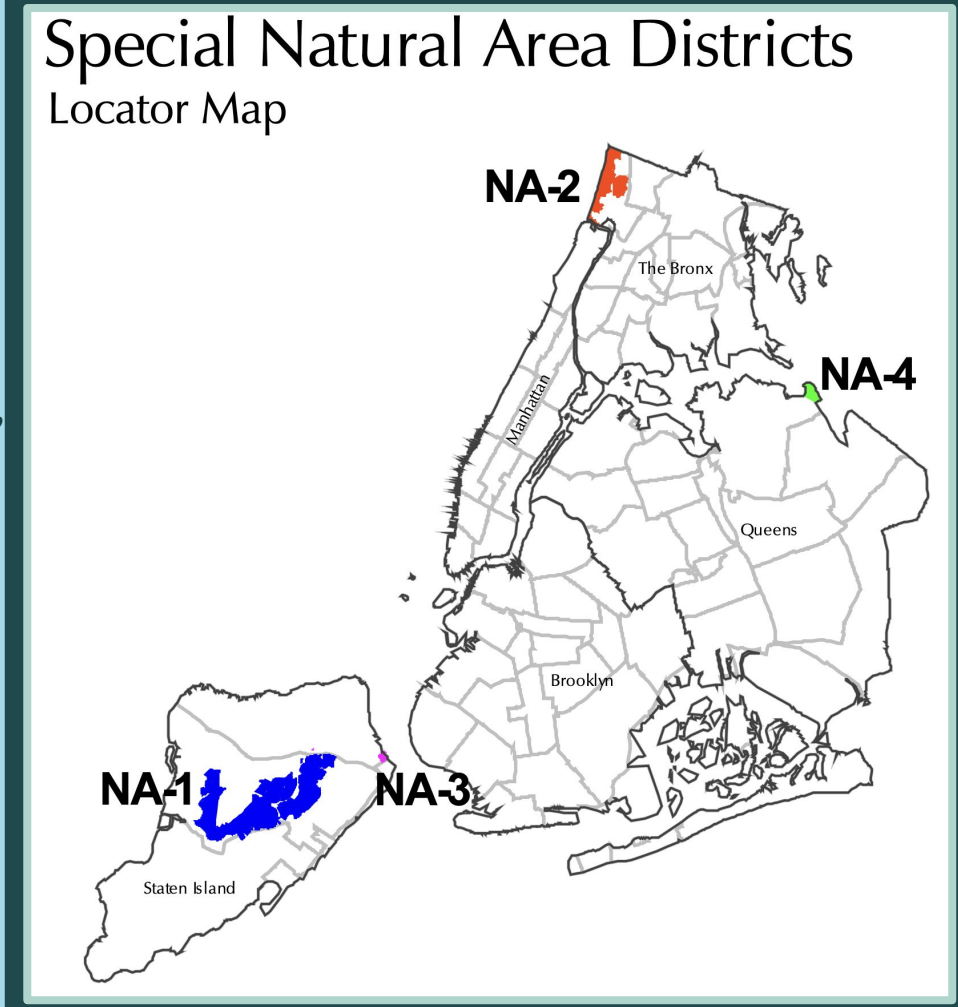
Karl Muller

Senior Software Engineer, Reliability & AI



- Infrastructure & Reliability Engineer at Boomi, Integration & Data Activation Platform
- Worked on RL in Gaming & ML in Aviation
- Forest Project Intern at Wave Hill, a private park in the Bronx
- **Today:** RL with Special Natural Area Districts (SNADs)

Contact
km2262@cornell.edu



Benefits for the DOB

Smarter SNAD Enforcement for the Department of Buildings

Smarter inspector dispatch

replaces ad-hoc triage with a data-driven policy, so inspector time goes to the highest yield complaints

Seasonal pre-positioning

agent learns Spring/Summer is peak risk (61.4% of complaints), enabling proactive resource allocation before violations escalate

Recurrent violator detection

−8 defer penalty encodes that serial violators cause irreversible environmental damage so the agent learns not to let them slip through

Transparent & tunable policy

Feature attribution shows why each complaint was flagged; the Pareto frontier gives managers a cost vs enforcement dial to match staffing and season

Analyzing Six Years of SNAD Complaints

- Source: NYC Open Data - 311 Service Requests, filtered to Problem Type == "Special Natural Area District (SNAD)" from Jan 2020 to Mar 2026
- 290 complaints (283 closed, 7 open), All records share one descriptor: "Illegal Tree Removal / Topological Change in SNAD"
- Geography: 63 zip codes, 37 community boards, dominated by Staten Island (63%), Queens (15%), Bronx/Riverdale (12%)
- Enforcement rate: 18.6% (54/290) [Violation Issued + ECB/OAH Notice + Stop Work Order]
- Seasonality: Spring + Summer = 61.4% of all complaints
- Backtest hold out: 87 complaints filed 2024–2026 (used for out of sample evaluation only)



Building & Improving the RL Agent

Markov Decision Process (MDP) Formulation

- **State space:** 128 states = 4 neighborhoods × 4 complaint levels × 2 recurrence flags × 4 seasons

- **Actions:** Defer / Inspect / Prioritize

- **Rewards:** sampled from empirical enforcement probabilities with seasonal (1.6×) and recurrence (1.4×) multipliers; –8 penalty for deferring recurrent violators

- **Algorithms:** Q-Learning (off-policy), SARSA (on-policy), DQN (9-dim continuous state, 2×64 ReLU, replay buffer)

Adding Interpretability & Flexibility

- **REINFORCE + Learned Baseline:** Monte Carlo policy gradient with advantage normalization (z-score); baseline network trained concurrently to reduce variance

- **Shapley Feature Attribution:** Exact Shapley values over 5 features (2^5 = 32 coalitions) on baseline network V(s)

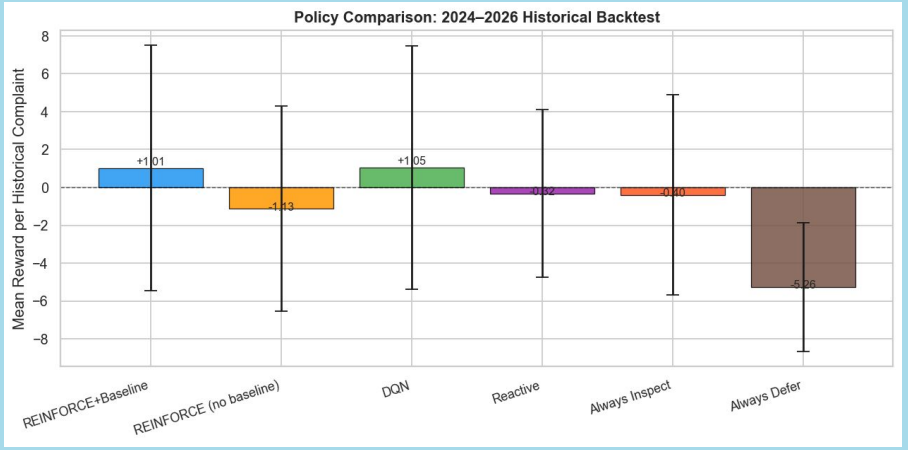
- **Pareto Frontier:** 5 reward variants re-trained and evaluated on Inspector Cost vs. Violation Remediation Rate

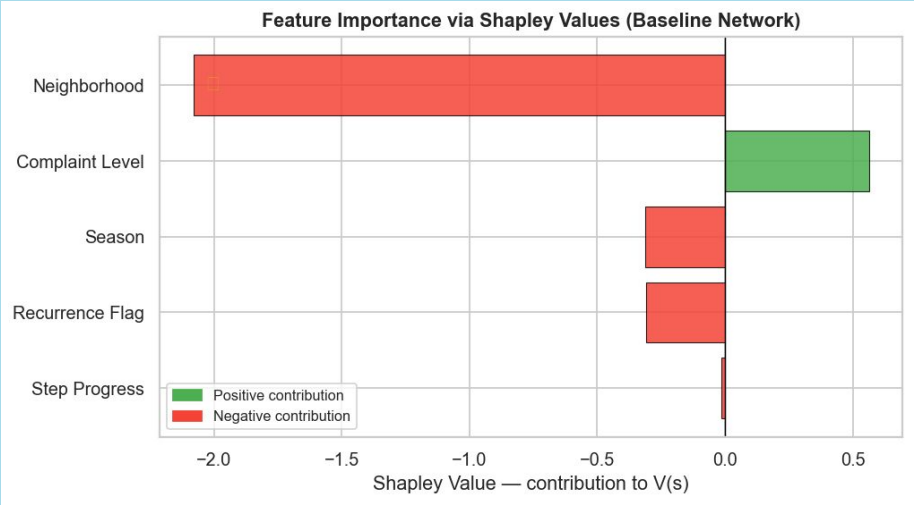
How the Agent Performed on Real Complaints

Backtest Data: 87 held out complaints (2024–2026)
- *DQN & REINFORCE + Baseline:* dispatching 85/87 complaints & achieving 7 enforcements
- *Reactive heuristic:* 52 / 87 dispatches, 5 enforcements
- *Inspect everything:* 87/87 dispatches, 7 enforcements, wasted dispatch effort

What the agent learned on its own
- *Neighborhood:* dominant signal, geography matters most
- *Complaint Level:* more activity means more real violations likely
- *Season & recurrence*: contribute equally, peak seasons often overwhelm capacity

Tuning dial / Pareto trade off
- *Balanced:* best agent approach, moderate dispatch cost, solid enforcement yield
- *Aggressive:* send inspectors everywhere, maximizes violations caught but maximizes cost





Thank you